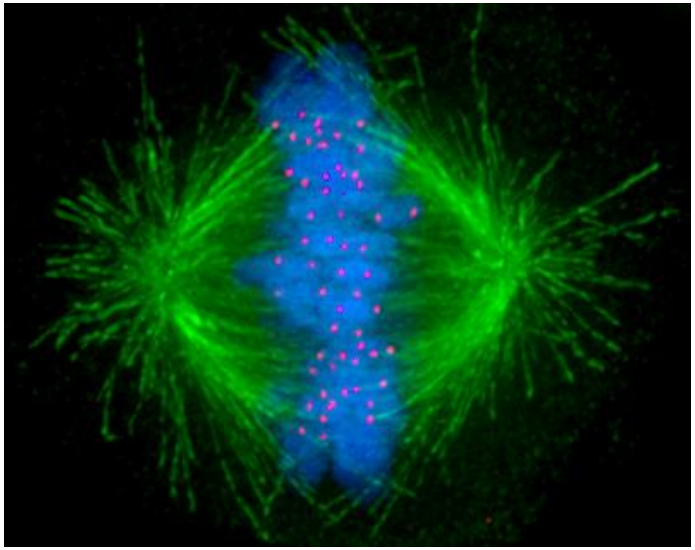
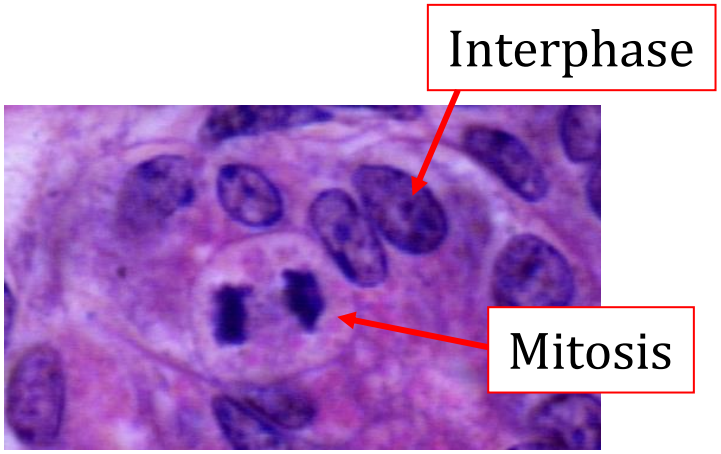
The cell cycle and mitotic cell division

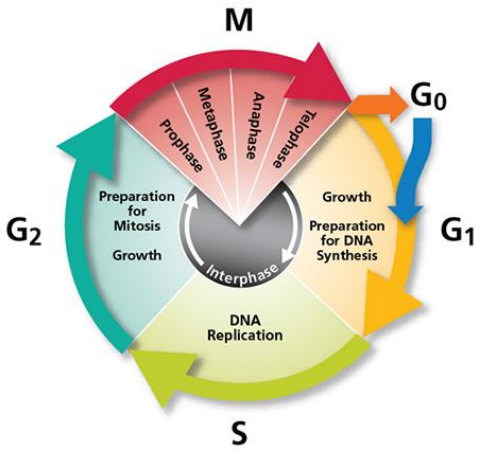


OVERVIEW

- Phases and events of the cell cycle.
- Cell cycle control at the checkpoints.
- Mitotic cell division.

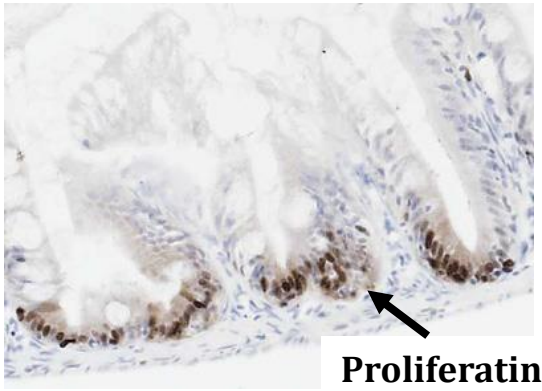
Phases and events of the cell cycle





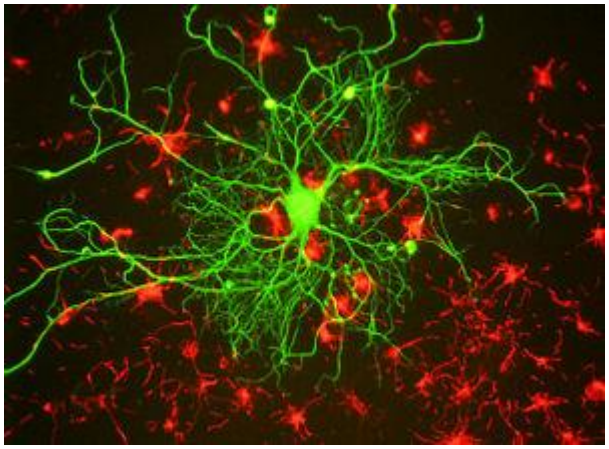
- **Interphase** (23h)
  - **G1** (11h) → Regular metabolic functions
  - **S** (8h) → -DNA synthesis -Histone synthesis
  - **G2** (4h) → Preparation for mitosis
- **Mitosis** (1h)

Cell cycle in cells within a specific tissue

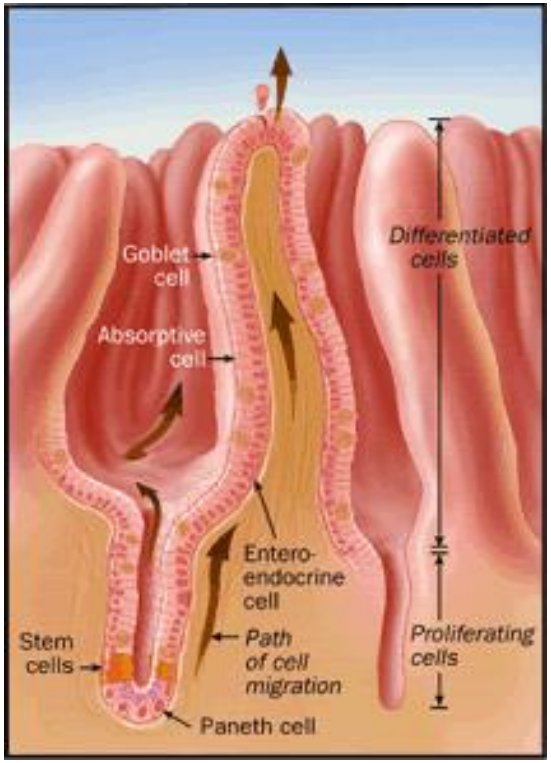


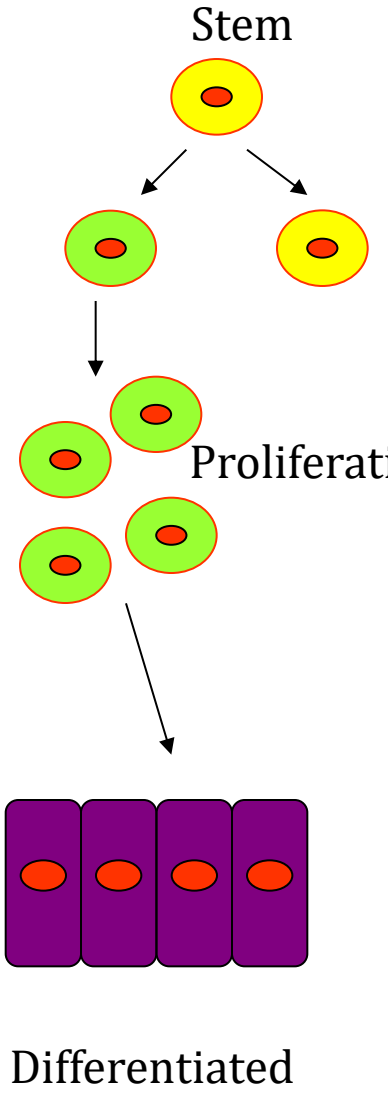
Proliferating cells

Proliferation vs. differentiation



Neurons

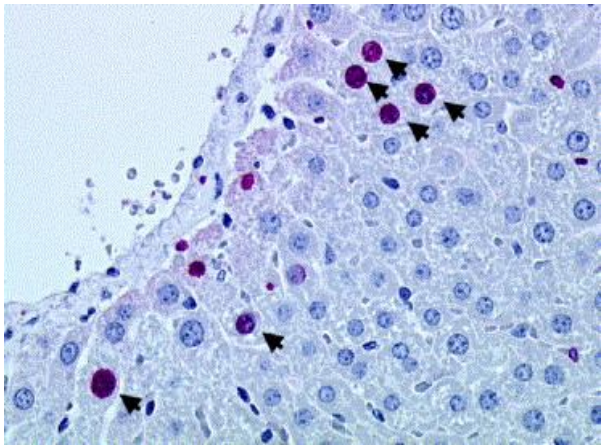




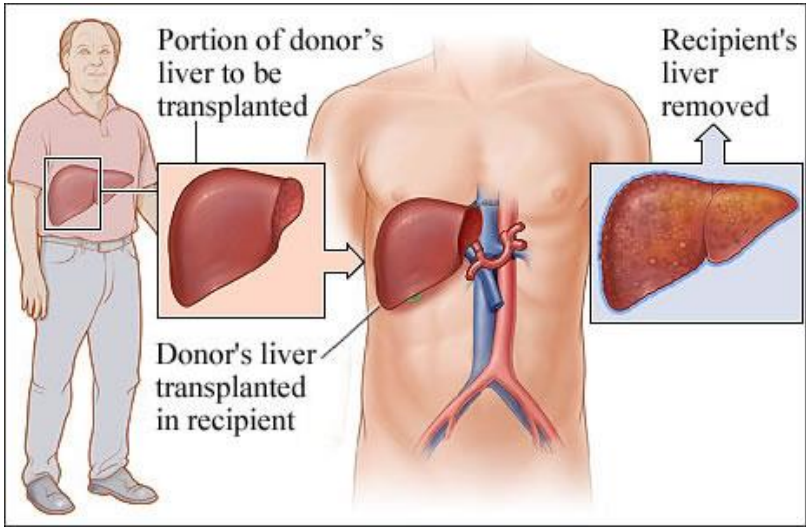
The G0 phase

Self-regeneration ability of the liver

Proliferating cells



Living-donor liver transplant



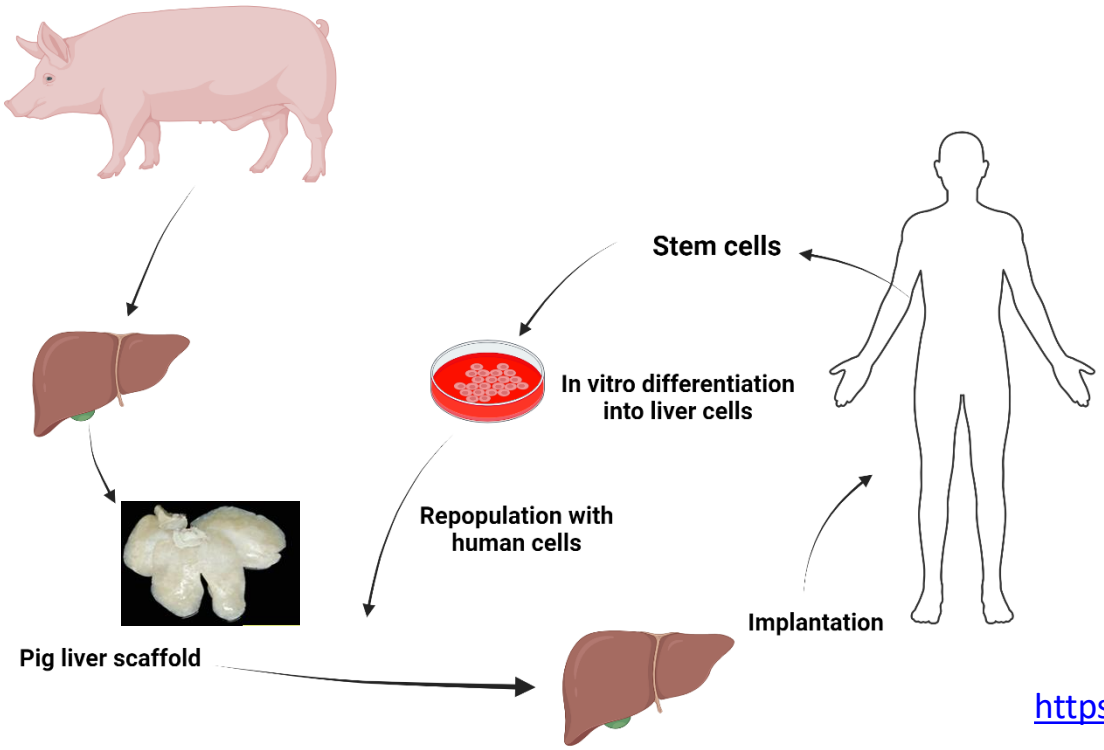
-Why are some stem cells able to regenerate injured tissues and others don't?

-How can we control this process? ⟶ Implications for tissue engineering and regeneration, transplants or cancer…

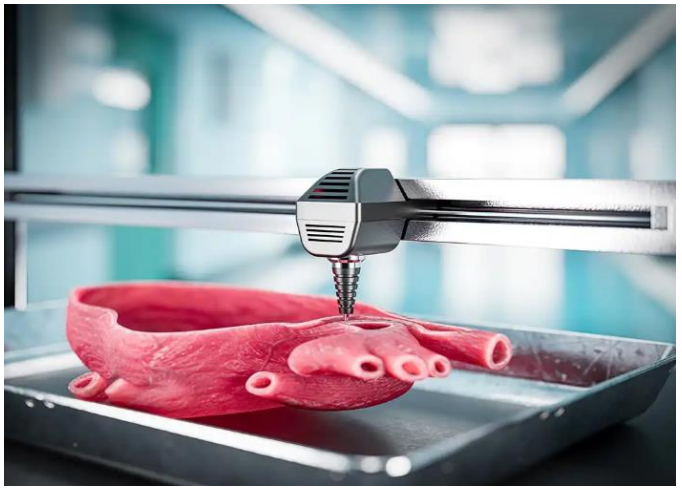
-Do we know enough about the cell cycle?

Examples of current research on tissue engineering

Creating organs from stem cells and scaffolds



Organ 3D bioprinting



Some interesting videos

https://www.youtube.com/watch?v=8l6ib9HitJ0

https://www.youtube.com/watch?v=V0rIP_u1JPQ

Take home message: more knowledge on how to control cell homing, growth and homeostasis is needed

Article

Gene-modified pig-to-human liver xenotransplantation

https://doi.org/10.1038/s41586-025-08799-1
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Accepted: 18 February 2025
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Kai-Shan Tao[1,16], Zhao-Xu Yang[1,16], Xuan Zhang[1,16], Hong-Tao Zhang[1,16], Shu-Qiang Yue[1], Yan-Ling Yang[1], Wen-Jie Song[1], De-Sheng Wang[1], Zheng-Cai Liu[1], Hai-Min Li[1], Yong Chen[1], Rui Ding[1], Shi-Ren Sun[2], Ming Yu[3], Ji-Peng Li[4], Wei-Xun Duan[5], Zhe Wang[6], Jing-Wen Wang[7], Jia-Yun Liu[8], Min-Wen Zheng[9], Xi-Jing Zhang[10,11], Wen Yin[12], Wei-Jun Qin[13], Dong-Mei Bian[14], Lin Li[1], Min Li[1], Zhi-Bin Lin[1], Hao Xu[1], Dan Wei[1], Hong Zhang[1], Juan-Li Duan[1], Deng-Ke Pan[15], Hai-Long Dong[12] ✉, Lin Wang[13] ✉ & Ke-Feng Dou[1] ✉

The shortage of donors is a major challenge for transplantation; however, organs from genetically modified pigs can serve as ideal supplements[1,2]. Until now, porcine hearts and kidneys have been successively transplanted into humans[3–7]. In this study, heterotopic auxiliary transplantation was used to donate a six-gene-edited pig liver to a brain-dead recipient. The graft function, haemodynamics, and immune and inflammatory responses of the recipient were monitored over the subsequent 10 days. Two hours after portal vein reperfusion of the xenograft, goldish bile was produced, increasing to 66.5 ml by postoperative day 10. Porcine liver-derived albumin also increased after surgery. Alanine aminotransferase levels remained in the normal range, while aspartate aminotransferase levels increased on postoperative day 1 and then rapidly declined. Blood flow velocity in the porcine hepatic artery and portal and hepatic veins remained at an acceptable level. Although platelet numbers decreased early after surgery, they ultimately returned to normal levels. Histological analyses showed that the porcine liver regenerated capably with no signs of rejection. T cell activity was inhibited by anti-thymocyte globulin administration, and B cell activation increased 3 days after surgery and was then inhibited by rituximab. There were no significant peri-operative changes in immunoglobulin G or immunoglobulin M levels. C-reactive protein and procalcitonin levels were initially elevated and then quickly declined. The xenograft remained functional until study completion.

Pig Kidney Recipient Returns Home After Transplant Breakthrough at NYU Langone Health

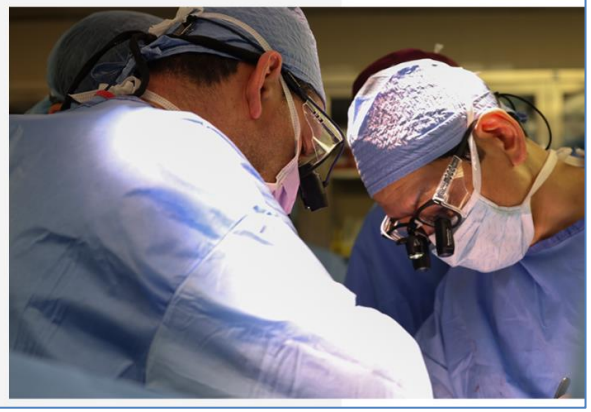


Home - News - Press Release

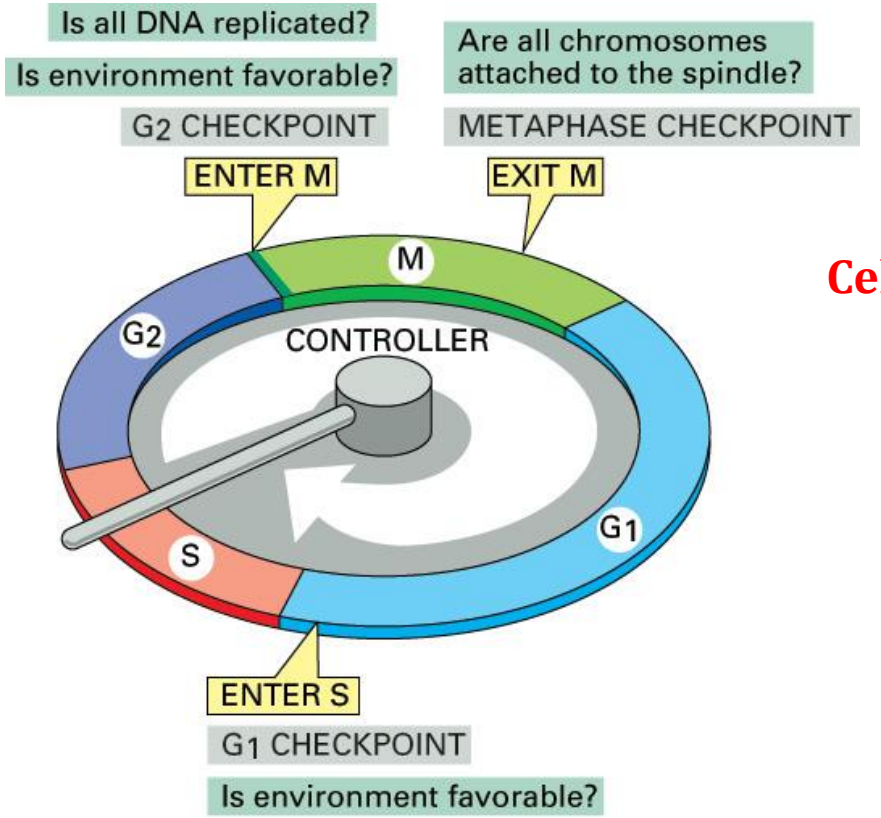
PRESS RELEASE · 5 MINUTE READ · MAR | 21 | 2024

World's First Genetically-Edited Pig Kidney Transplant into Living Recipient Performed at Massachusetts General Hospital

Brandon Chase · bchase7@mgb.org



Cell cycle control at the checkpoints



Restriction Point ("Start")

Cell commitment at the Restriction Point



Proteins that control the checkpoints:

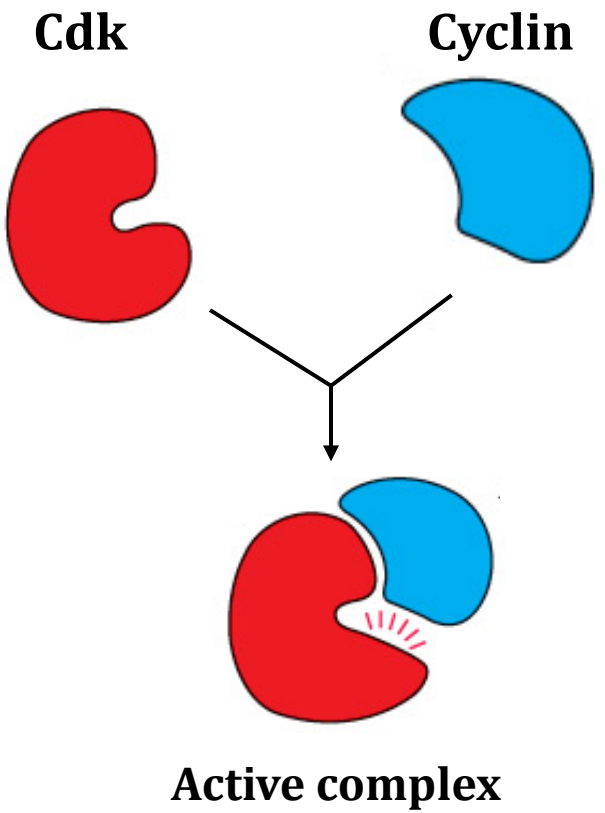
Cdk-cyclins complexes

Cdk (**Cyclin-dependent kinases**)

-Phosphorylate other proteins, when active.

-Activation requires binding to cyclins.

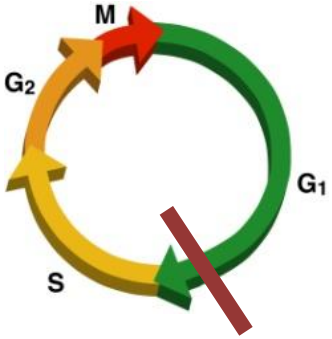
-Constitutive expression.



Cyclins:

- Bind to Cdk to control their activity.
- Cyclic expression.

Control at the restriction point



R. point

Growth factor

membrane

Receptor

Intracellular messengers

Cdk4

Cyclin D

P

P

Nucleus

Retinoblastoma

(Rb)

E2F

DNA synthesis

Histone synthesis

Transcription of genes

Cyclin D gene

Alteration of the system

Inactivating mutations in the Rb gene?

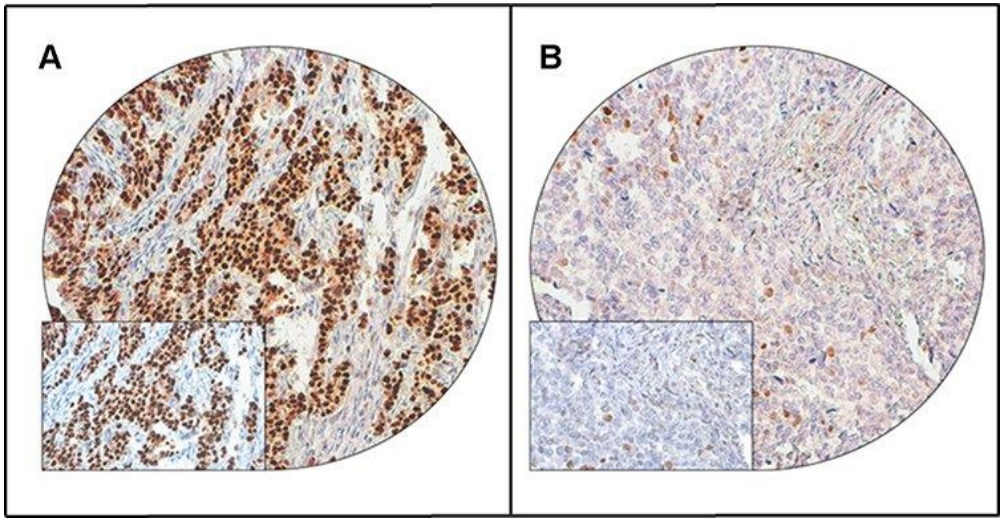
Abnormal increased levels of cyclins?



Retinoblastoma and other tumors

Tumor specimen with high cyclin D levels

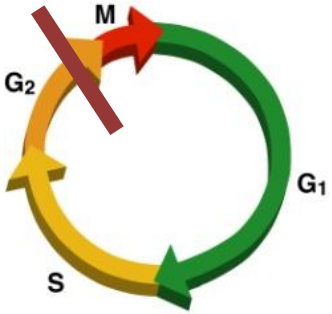
Tumor specimen with low cyclin D levels



Worse prognosis

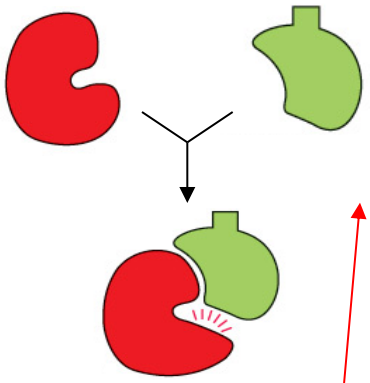
Control at the G2/M point

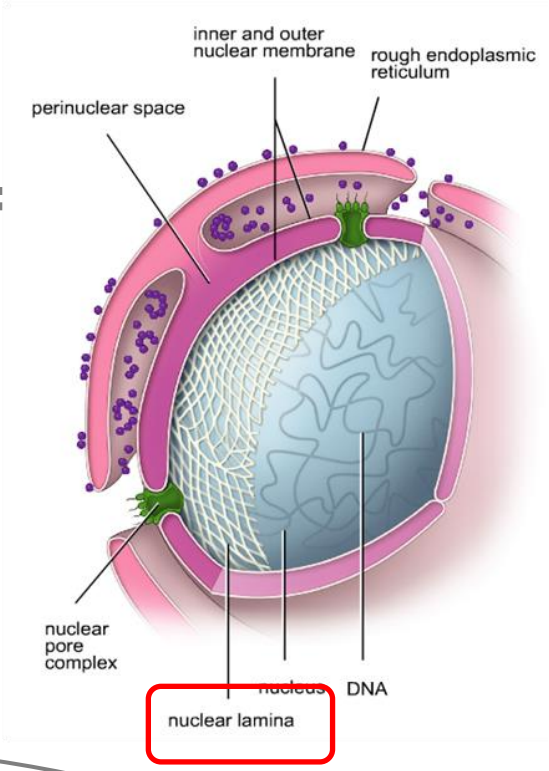
G2/M point



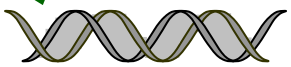
membrane

Cdk1 Cyclin B





-Correct DNA replication
-Correct size
-Favorable external conditions



Cyclin B gene

P Proteins involved in:
-Nuclear lamina (break-down)
-Chromosome condensation
-Spindle formation

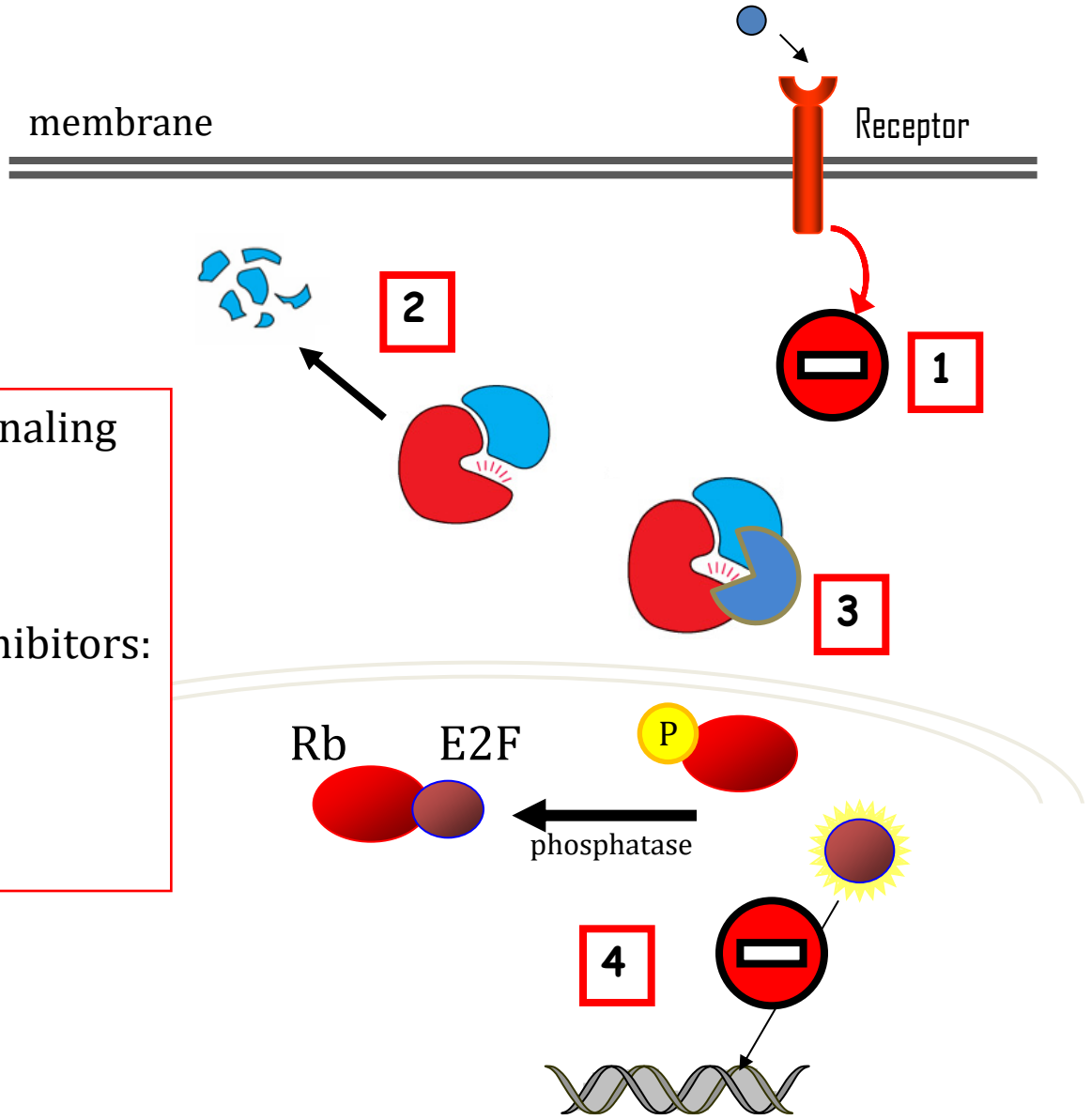
Inactivation of the Cell Cycle

1. Inhibition of intracellular signaling

2. Cyclin degradation

3. Production of Cdk/cyclin Inhibitors:
   p21

4. De-phosphorylation of Rb



Mitotic cell division

https://www.youtube.com/watch?v=8uzHTKdv_Sw

